QUALITY ASSURANCE PLAN (QAP)

Roaya

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Smart City Transformation

Project title	Roaya – Smart City Transformation
Team members	Project Manager, Planning Manager, Technical Manager, Finance Manager, Risk & Quality Manager, Stakeholder Manager
Date	[Month Year]

1. Purpose and Scope

This Quality Assurance Plan explains how the team will make sure the Roaya project meets all required standards of quality, safety, security and citizen satisfaction. It covers the whole project, from assessment and design to installation, testing, the pilot and city-wide deployment.

2. Quality Objectives

- All smart systems work reliably, with at least 99% uptime after launch.
- All equipment meets safety, electrical and cybersecurity standards.
- Citizen data is protected and follows data protection law.
- Achieve at least 85% citizen satisfaction during the pilot.
- Complete the project on time (December 2028) and within the EGP 550M budget.

3. Quality Standards and Criteria

Area	Standard / Criteria	Measurement Method
Infrastructure	Networks, sensors and cameras are installed safely and work correctly	Site inspection and installation checklist
Equipment	Devices meet safety, quality and performance requirements	Supplier certificates and testing
Software & apps	City app and control center are fast, stable and easy to use	System testing and user testing
Cybersecurity & data	Systems are secure and citizen data is protected	Security testing and data protection review
Citizen services	Online services are available and easy to access	Citizen feedback and service reports

4. Roles and Responsibilities

Role	Quality responsibilities
Project Manager	Supervises all phases and gives final approval once all quality criteria are met
Risk & Quality Manager	Leads quality checks, security testing and the Quality Log
Technical Manager	Makes sure infrastructure and systems meet technical standards
Planning Manager	Tracks quality milestones in the schedule
Finance Manager	Checks supplier quality before purchase and keeps costs within budget
Stakeholder Manager	Collects citizen feedback and shares quality updates
Project Sponsor	Reviews and approves the final outcomes

5. Quality Assurance Activities

Activity	Purpose	Timing / Frequency	Responsible Person
Design review	Check the architecture and governance design meet standards	Before installation (Phase 3)	Technical Manager

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Supplier evaluation	Verify the quality and reliability of suppliers and contractors	Before purchase	Finance Manager
Installation inspection	Check networks, sensors and cameras are installed correctly	During Phase 4	Technical Manager
System and security testing	Test the control center, app and cybersecurity	Before the pilot (Phase 5)	Risk & Quality Manager
Staff training	Make sure city staff know the systems and quality procedures	Before launch	Stakeholder Manager
Citizen feedback	Collect opinions on the new services	During and after the pilot	Stakeholder Manager

6. Quality Control Process

1. Use inspection checklists for design, installation and systems.
2. Test all equipment and systems before they go live.
3. If any issue is found, record it in the Quality Log and assign a corrective action immediately.
4. Run a final inspection and test run before the pilot and before city-wide launch.
5. The Project Manager gives final approval once all criteria are met.

7. Documentation and Reporting

- Keep a Quality File with supplier certificates, inspection reports, test results and feedback forms.
- Submit monthly quality updates to the project sponsor.
- Record all non-conformities and the actions taken to fix them.
- Prepare a final Quality Report summarizing all quality activities and outcomes.

8. Continuous Improvement

- After launch, collect citizen feedback and analyze the results.
- Discuss lessons learned at the end of each phase.
- Identify areas for improvement (for example service speed, app design or system reliability).
- Update the quality plan for future smart city projects.